

Differentially Private Preference Data Synthesis for Large Language Model Alignment

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Abstract

Preference alignment is a central post-training procedure for large language models (LLMs), in which model outputs are optimized to conform to human preference signals. Direct post-training on real preference corpora, however, creates privacy exposure because such corpora often contain sensitive user prompts, candidate responses, and annotator judgments. We propose **DPPrefSyn**, an algorithmic framework for generating differentially private (DP) synthetic preference data for privacy-preserving preference alignment. DP-PrefSyn is specified through the Bradley–Terry preference model and the geometric structure of pairwise preference representations. The method first estimates a preference model from private data under formal differential privacy guarantees, and then combines this model with public prompts to synthesize preference pairs. To represent heterogeneous human preferences, DP-PrefSyn uses a shared linear structure across per-cluster reward models; to improve estimation in high-dimensional embedding spaces, it applies DP Principal Component Analysis (DP-PCA). Across extensive experiments, DPPrefSyn attains competitive alignment utility under strong DP constraints. These results identify synthetic preference data as a technically viable substitute for direct private-data post-training in privacy-preserving LLM alignment. To the best of our knowledge, this is the first work to generate DP synthetic preference data for LLM alignment. Our code will be released upon acceptance.

1. Introduction

Preference alignment methods, including RLHF (Stiennon et al., 2020) and DPO (Rafailov et al., 2024), constitute

standard post-training mechanisms for calibrating LLM behavior to human expectations. These methods are typically parameterized by pairwise preference datasets in which annotators compare two candidate responses for a prompt and identify the response that better satisfies the task specification. The resulting comparisons are used to fine-tune LLMs so that preferred responses receive higher implicit or explicit preference scores than dispreferred responses (Ziegler et al., 2019; Ouyang et al., 2022; Bai et al., 2022). The empirical effectiveness of this paradigm has supported deployment in chat assistants (Achiam et al., 2023), mathematical reasoning systems (Shao et al., 2024), and code generation models (Shen et al., 2023).

Human-preference alignment also induces a nontrivial privacy problem: the underlying datasets frequently contain real user instructions, model responses, and human feedback. Prompts can reveal health information, identity attributes, or other sensitive context, while feedback can encode private beliefs, preferences, and behavioral patterns (Li et al., 2023; Yu et al., 2024). Prior work (Chowdhury et al., 2024; Yu et al., 2024; Wu et al., 2023a) has used differential privacy (DP) (Dwork et al., 2006b) to mitigate these risks and has reported promising empirical behavior. Nevertheless, many existing mechanisms privatize only a subset of the preference record, such as the user prompt (Yu et al., 2024) or the preference label (Chowdhury et al., 2024; Zhang et al., 2025), rather than the prompt, responses, and label jointly. This partial protection leaves residual attack surfaces because sensitive attributes may still be inferred from the unprotected fields. In addition, several methods are tied to a particular post-training algorithm, such as RLHF (Wu et al., 2023a), and therefore do not directly transfer to newer objectives such as DPO. Finally, because human preference annotation is expensive and difficult to scale, direct private-data fine-tuning is often limited by the size of the private corpus and may produce insufficient alignment utility. These gaps motivate the central question of this work:

Is it possible to achieve superior preference alignment while ensuring thorough privacy protection and maintaining compatibility with emerging alignment methodologies?

We answer this question affirmatively through the following

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technical contributions:

- We introduce DPPrefSyn (Figure 1), an algorithm for producing DP synthetic preference datasets for LLM alignment. The construction is grounded in the Bradley–Terry model and in the geometry of pairwise preference embeddings. It has three coupled design elements: 1) *Capturing diverse human preferences via shared linear structure of rewards and clustering*. The Bradley–Terry model with a linear reward parameterization induces a clustering criterion based on embedding differences between concatenated prompt–preferred and prompt–dispreferred pairs, allowing each cluster to be represented by a shared reward function. 2) *Improving learning efficiency through DP-PCA*. To reduce sample inefficiency and clustering instability in high-dimensional embeddings, DPPrefSyn applies DP-PCA before clustering and reward estimation, preserving the principal preference-relevant directions under DP. 3) *Saving privacy budgets by using public prompts*. DPPrefSyn generates candidate responses from public prompts and uses privatized reward models to select preference pairs, thereby allocating the privacy budget to private preference modeling rather than prompt synthesis.
- We provide a formal privacy analysis establishing that DPPrefSyn satisfies (ϵ, δ) -DP. The analysis allocates ϵ_0 to DP-PCA (Amin et al., 2019) for dimensionality reduction and ϵ_1 to DP-KMeans (Su et al., 2016) for clustering; the residual budget is assigned to per-cluster reward-model training via DP-SGD. We use the Privacy Random Variable (PRV) accountant (Gopi et al., 2021) to compose the DP-SGD privacy loss tightly. By DP post-processing, the generated synthetic dataset can subsequently be reused with multiple alignment algorithms and downstream LLMs without additional privacy expenditure.
- We evaluate DPPrefSyn on standard alignment benchmarks, including question answering tasks from OpenAssistant (Köpf et al., 2023) and Anthropic-HH (Bai et al., 2022), and the TL;DR summarization task (Stiennon et al., 2020). The results show that DPPrefSyn preserves formal privacy guarantees while maintaining strong utility across task families. For example, at $\epsilon = 2$, DPO fine-tuning of Pythia-2.8B on DPPrefSyn data obtains a GPT-4o win rate of 56.48% on Anthropic-HH, compared with 37.02% for fine-tuning on real data. The synthetic dataset is also compatible with multiple preference-optimization procedures, including DPO and RLHF-style training, addressing the method-specificity of several prior privacy-preserving post-training approaches.

2. Related Work

Differentially Private Alignment of LLMs. Wu et al.

(2023a) formulate a DP alignment framework by adapting PPO to the differentially private setting. This construction, however, is *limited to PPO* and therefore does not directly cover the broader and evolving class of preference-optimization objectives. Chowdhury et al. (2024) analyze reward estimation from preference-based feedback under label-DP (Ghazi et al., 2021), and Zhang et al. (2025) propose AUP-RLHF as a user-level label-DP framework for RLHF. These methods protect preference labels, but they *do not address the sensitivity of prompts or responses themselves*. They are also typically *constrained by the limited private user preference data*, which can restrict alignment quality because human preference annotations are expensive to collect. Yu et al. (2024) instead protect sensitive user *instructions* by generating DP synthetic *instructions* that replace real instructions during annotation and fine-tuning.

Outside preference alignment, a separate literature studies private LLM fine-tuning on conventional labeled examples rather than pairwise comparisons (Yu et al., 2021; Li et al., 2021; Chen et al., 2024; Tang et al., 2024; Zhang et al., 2023; He et al., 2022). Another line protects prompts used for in-context learning, where privacy constraints apply at inference-time context construction rather than post-training data synthesis (Duan et al., 2023; Tang et al., 2023; Gao et al., 2024; Hong et al., 2023; Wu et al., 2023b).

Differentially Private Synthetic Text Generation. DP-PrefSyn is also related to DP synthetic text generation. Existing methods can be organized into two representative categories. The first category fine-tunes a language model on private data with DP and then samples synthetic text from the privatized model (Yue et al., 2023; Mattern et al., 2022; Miresghallah et al., 2023; Carranza et al., 2023; Yu et al., 2024; Wang et al., 2024; Ochs & Habernal, 2024; Tan et al., 2025; Carranza et al., 2024). For example, Yu et al. (2024) privately fine-tune LLaMA on private instruction data and use the resulting generator to produce text for supervised instruction tuning.

Because DP fine-tuning is computationally costly and may be infeasible for non-public language models, a second category generates DP synthetic data through LLM APIs (Lin et al., 2023; Xie et al., 2024; Wu et al., 2024; Hou et al., 2024). These methods typically query an external generator and then select or refine candidate samples according to a privacy-preserving criterion measuring similarity to private data. For instance, Xie et al. (2024) propose Aug-PE, which iteratively updates LLM-generated samples using embedding-space similarity to the private corpus. *Our work follows this API-based synthesis perspective but instantiates it for pairwise preference alignment.*

Diversity in Human Preferences. Recent work emphasizes that human preferences are heterogeneous rather than monolithic (Denton et al., 2021; Aroyo et al., 2023a;b;

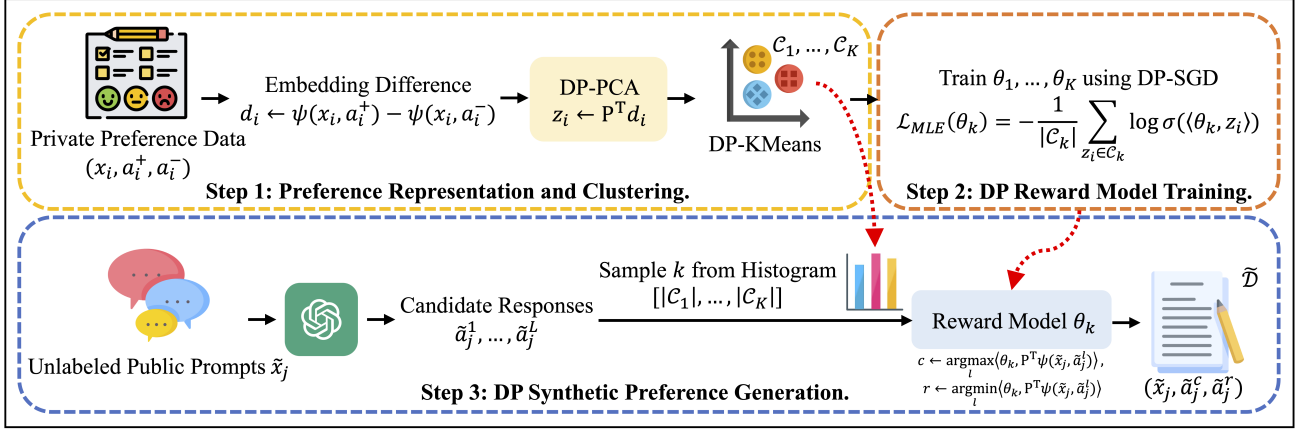


Figure 1. Overview of DPPrefSyn. The algorithm constructs DP synthetic preference data in 3 stages: 1) it encodes preference records as embedding-difference vectors and clusters them using DP-PCA and DP-KMeans, 2) it estimates a DP reward model within each cluster using DP-SGD, and 3) it samples reward functions from the DP-protected cluster distribution to synthesize preference pairs from public prompts.

Chakraborty et al., 2024). Sources of heterogeneity include sociodemographic attributes (e.g., race, gender, age), personal beliefs and biases, differences in domain expertise, and ambiguity in natural-language specifications (Sandri et al., 2023; Vogels, 2021). This observation has motivated personalized alignment methods that attempt to represent the values of distinct user groups (Lee et al., 2024b; Poddar et al., 2024; Singh et al., 2025). For example, Singh et al. (2025) propose a meta-learning procedure in which an LLM adapts to a user from a small set of preference examples. DPPrefSyn generates DP synthetic preference data that captures the underlying diversity of human feedback; as shown in Section 5.3, explicitly modeling this diversity improves alignment performance relative to variants that collapse preferences into a single group.

3. Problem Definition

Preference Alignment of LLMs. We study preference alignment for an LLM given pairwise human feedback represented by a dataset $\mathcal{D} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$. Each record contains a prompt x_i and two candidate responses a_i^+ and a_i^- , with a_i^+ denoting the annotator-preferred response.

Following prior work on reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) (Ouyang et al., 2022; Zhu et al., 2023; Rafailov et al., 2024; Liu et al., 2023), we parameterize the preference probability using the Bradley–Terry model (Bradley & Terry, 1952):

$$\mathbb{P}[a_i^+ \succ a_i^- \mid x_i] = \frac{\exp(r^*(x_i, a_i^+))}{\exp(r^*(x_i, a_i^+)) + \exp(r^*(x_i, a_i^-))} = \sigma(r^*(x_i, a_i^+) - r^*(x_i, a_i^-)),$$

where $a_i^+ \succ a_i^-$ denotes the event that a_i^+ is preferred to a_i^- ,

$r^*(x, a)$ is the latent reward assigned to a prompt-response pair, and $\sigma(z) = 1/(1 + e^{-z})$ is the sigmoid link function.

This probabilistic formulation supports both RLHF, which estimates an explicit reward model from preference comparisons, and DPO, which induces an implicit reward through the log-ratio between the learned policy and a reference policy.

Differentially Private Preference Data Synthesis. The privacy objective is to protect the private preference dataset $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ against an adversary attempting to recover or infer information about individual prompts, responses, or preference labels from the outputs of a fine-tuned LLM. We therefore construct a DP synthetic preference dataset $\tilde{\mathcal{D}} = \{(\tilde{x}_j, \tilde{a}_j^+, \tilde{a}_j^-)\}_{j=1}^m$ that retains the preference information needed for alignment while satisfying a formal privacy guarantee. By the post-processing property of DP, $\tilde{\mathcal{D}}$ can be supplied to different alignment algorithms and LLM architectures without additional privacy loss.

We use the following standard definition of (ϵ, δ) -differential privacy.

Definition 3.1 ((ϵ, δ) -Differential Privacy (Dwork et al., 2006a)). A randomized algorithm $\mathcal{A}$ is (ϵ, δ) -differentially private if for any two neighboring inputs $\mathcal{D}$ and $\mathcal{D}'$ that differ by a single entry and any set $\mathcal{S}$ of possible outputs: $\mathbb{P}[\mathcal{A}(\mathcal{D}) \in \mathcal{S}] \leq e^\epsilon \mathbb{P}[\mathcal{A}(\mathcal{D}') \in \mathcal{S}] + \delta$.

Diverse User Preferences and Linear Reward Structure. Human preferences in $\mathcal{D}_{\text{priv}}$ are generally not homogeneous (Chakraborty et al., 2024; Bakker et al., 2022; Kovač et al., 2023; Jang et al., 2023; Rame et al., 2023; Ji et al., 2023). Annotators may weight factual accuracy, politeness, creativity, clarity, and other response attributes differently, producing heterogeneous and sometimes conflicting comparison signals. A synthetic dataset $\tilde{\mathcal{D}}$ should therefore preserve the

distributional structure of these heterogeneous preferences rather than only their aggregate majority trend.

To represent this structure, we assume a linear latent reward model (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023). Specifically, we adopt

$$r_\theta(x, a) = \langle \theta, \phi(x, a) \rangle,$$

where $\phi(x, a) \in \mathbb{R}^d$ is a *shared* feature representation of the prompt-response pair and $\theta \in \mathbb{R}^d$ is a *user-dependent* preference parameter.

This parameterization provides a controlled trade-off between *expressiveness* and *tractability*. Although users may express heterogeneous preferences, their reward functions can share semantic basis features in the embedding space, represented by $\phi(x, a)$, while differing in the weights assigned to those features, represented by θ . The resulting model captures shared structure and user- or cluster-level variation with a sample-efficient parameterization, which is important under limited private data and DP noise. The empirical study in Section B.6 further shows that this linear reward structure captures preference signals competitively relative to a fully fine-tuned GPT-2 reward model.

4. Proposed Method

4.1. Key Components of the Algorithm Design

We first specify the algorithmic components that resolve the main technical obstacles in differentially private preference-data synthesis.

Leveraging Linear Reward Structure for Preference Clustering. The linear reward model implies

$$\mathbb{P}[a_i^+ \succ a_i^- \mid x_i] = \sigma(\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle).$$

Consequently, for a tuple (x_i, a_i^+, a_i^-) , the event $a_i^+ \succ a_i^-$ is associated with $\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle \gg 0$; equivalently, the preference vector θ is highly aligned with the feature-difference vector $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$. If two annotators have similar preference parameters, the corresponding feature-difference vectors for their rated pairs should concentrate along similar directions in the embedding space. DPPrefSyn therefore clusters preference records according to $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$ so that records assigned to the same cluster can be approximated by a common reward vector θ . In Section B.9, we empirically verify that these clusters correspond to interpretable preference styles.

Adopting DP-PCA to Obtain Effective Features. A natural instantiation of $\phi(x, a)$ is the embedding of the associated prompt-response pair, because such embeddings encode semantic and stylistic attributes relevant to human comparisons. These embeddings are typically high-dimensional (e.g., 1024 dimensions), which increases sample complexity

and destabilizes clustering when each cluster contains only a limited number of private records. We therefore apply differentially private PCA (DP-PCA) (Amin et al., 2019; Liu et al., 2022) to obtain a lower-dimensional representation before clustering and reward-model estimation. This projection improves statistical efficiency and numerical stability while preserving the dominant preference-relevant directions under DP.

Utilizing Public Prompts to Save DP Budget. Jointly synthesizing prompts, preferred responses, and dispreferred responses would allocate privacy budget across all fields and can substantially reduce utility. DPPrefSyn instead uses public prompts and spends the privacy budget only on estimating private preference structure over responses. For each public prompt, an LLM generates multiple candidate responses; a DP-trained distribution of reward functions then selects the preferred and dispreferred candidates. Although the public prompt distribution may not match the private prompt distribution, the experiments show that the induced synthetic preference pairs preserve the private preference distribution sufficiently for downstream alignment.

4.2. DPPrefSyn Algorithm

We now define DPPrefSyn (Algorithm 1), which maps the private preference dataset $\mathcal{D}_{\text{priv}}$ and a public prompt set to a DP synthetic preference dataset. The procedure has three stages.

Step 1: Preference Representation and Clustering. For each preference record in $\mathcal{D}_{\text{priv}}$, DPPrefSyn constructs a pairwise representation using a public sentence embedding model ψ . Given (x_i, a_i^+, a_i^-) , the prompt is concatenated with each response to form $[x_i; a_i^+]$ and $[x_i; a_i^-]$. The algorithm then evaluates ψ on both texts and forms the signed embedding difference $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ (Line 3).

To privatize and reduce the representation dimension, DPPrefSyn applies DP-PCA (Amin et al., 2019) to $\{d_i\}_{i=1}^n$, producing a projection matrix $\mathbf{P} \in \mathbb{R}^{d \times p}$ under ϵ_0 -DP (Line 5). At a high level, DP-PCA (Amin et al., 2019) privately approximates the eigendecomposition of the empirical covariance matrix by estimating eigenvalues and eigenvectors under DP. Each difference vector is projected as $z_i \leftarrow \mathbf{P}^\top d_i$ (Line 6). DPPrefSyn then runs DP-KMeans (Su et al., 2016) on $\{z_i\}_{i=1}^n$ to partition records with similar preference directions into K clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$ under ϵ_1 -DP (Line 7). DP-KMeans (Su et al., 2016) can be viewed as a differentially private analogue of Lloyd’s algorithm.

Step 2: DP Reward Model Training. For each cluster $\mathcal{C}_k$, DPPrefSyn estimates a shared preference vector $\theta_k \in \mathbb{R}^p$ by minimizing the negative log-likelihood (Line 9):

$$\mathcal{L}_{\text{MLE}}(\theta_k) = -\frac{1}{|\mathcal{C}_k|} \sum_{z_i \in \mathcal{C}_k} \log \sigma(\langle \theta_k, z_i \rangle), \quad (1)$$

where z_i denotes the DP-PCA-projected embedding difference from Step 1. Each cluster-specific reward model is optimized with DP-SGD (Abadi et al., 2016) using noise multiplier σ_2 .

Step 3: DP Synthetic Preference Generation. DPPrefSyn computes the cluster-size histogram $\mathbf{h} = [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ (Line 11) and normalizes it to the sampling distribution $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ over clusters (Line 12).

For each public prompt $\tilde{x}_j$, the algorithm queries an LLM at a relatively high temperature to generate L candidate responses $\tilde{a}_j^1, \dots, \tilde{a}_j^L$ (Line 14). It then samples a cluster index $k \sim \mathbf{p}$ from the DP histogram (Line 15) and scores each candidate with the corresponding reward vector θ_k . Specifically, DPPrefSyn embeds each candidate prompt-response pair as $\psi(\tilde{x}_j, \tilde{a}_j^l)$, projects it using $\mathbf{P}$, and assigns reward $\langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$. The highest-scoring and lowest-scoring candidates are selected as the preferred and dispreferred responses, respectively: $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$, $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ (Lines 16-17). If the reward gap is below a quality threshold (e.g., < 0.5), the candidate tuple is discarded; otherwise, $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$ is inserted into $\tilde{\mathcal{D}}$. Repeating this operation over public prompts yields the DP synthetic preference dataset $\tilde{\mathcal{D}}$.

Algorithm 1 DPPrefSyn

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1: Input: Private dataset  $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ , public prompts  $\{\tilde{x}_j\}_{j=1}^m$ , embedding model  $\psi$ , number of clusters  $K$ , an LLM  $\text{LLM}(\cdot)$ , DP parameters  $\varepsilon_0, \varepsilon_1, \sigma_2$ .
2: for each  $(x_i, a_i^+, a_i^-)$  in  $\mathcal{D}_{\text{priv}}$  do
3:    $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ , where  $d_i \in \mathbb{R}^d$ 
4: end for
5:  $\mathbf{P} \leftarrow \text{DP-PCA}(\{d_i\}_{i=1}^n, \varepsilon_0)$ , where  $\mathbf{P} \in \mathbb{R}^{d \times p}$ 
6:  $\{z_i\}_{i=1}^n \leftarrow \{\mathbf{P}^\top d_i\}_{i=1}^n$ 
7:  $\mathcal{C}_1, \dots, \mathcal{C}_K \leftarrow \text{DP-KMeans}(\{z_i\}_{i=1}^n, K, \varepsilon_1)$ 
8: for each cluster  $\mathcal{C}_k$  do
9:   Train a linear reward model  $\theta_k$  on  $\mathcal{C}_k$  using DP-SGD with noise multiplier  $\sigma_2$  (Equation (1))
10: end for
11:  $\mathbf{h} \leftarrow [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ 
12:  $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ 
13: for each public prompt  $\tilde{x}_j$  do
14:   Generate responses  $\tilde{a}_j^1, \dots, \tilde{a}_j^L \leftarrow \text{LLM}(\tilde{x}_j)$ 
15:   Sample cluster index  $k \sim \mathbf{p}$ 
16:    $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
17:    $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
18:   Add  $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$  to  $\tilde{\mathcal{D}}$ 
19: end for
20: return  $\tilde{\mathcal{D}}$ .
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Privacy Analysis of DPPrefSyn. The privacy budget is divided across the private-data access operations: ε_0 is as-

signed to DP-PCA (Line 5), ε_1 is assigned to DP-KMeans (Line 7), and the remaining budget is used by DP-SGD for per-cluster reward-model training with noise multiplier σ_2 (Line 9). For the DP reward models, parallel composition applies because the clusters are disjoint; changing one training record affects at most one cluster, so the privacy cost is governed by the smallest retained cluster. DPPrefSyn enforces a known lower bound on cluster size by discarding clusters with too few samples. The Privacy Random Variable (PRV) accountant (Gopi et al., 2021) composes the per-step DP-SGD costs and verifies the residual guarantee $(\varepsilon - \varepsilon_0 - \varepsilon_1, \delta)$ -DP. All subsequent operations use public data or DP post-processing and therefore incur no additional privacy cost. We provide the full proof and privacy-accounting code in Section A.

Discussion on Prompt Distribution Shift. DPPrefSyn is designed to remain effective when private and public prompts follow different distributions. The heterogeneous preference structure is represented by θ_k , the parameter of the linear reward model, rather than solely by $\phi(x, a)$, the feature vector of an individual prompt-response pair. Intuitively, a user may issue prompts from a broad topic distribution while maintaining relatively stable response preferences. Thus, if the distribution over θ_k approximates the distribution in the private dataset, synthetic comparisons generated from public prompts can still encode the private preference patterns. DPPrefSyn implements this approximation by sampling a cluster index k from the DP histogram (Line 15) for each public prompt and then selecting response pairs with the associated reward model. Consequently, prompt-distribution mismatch does not by itself prevent the synthetic data from preserving the preference patterns learned from private data.

5. Experiments

Datasets. We evaluate DPPrefSyn on question-answering and summarization alignment tasks. For question answering, we use OpenAssistant (Köpf et al., 2023), which contains assistant-style conversations, and Anthropic-HH (Bai et al., 2022), which contains helpfulness- and harmlessness-oriented preference comparisons. For summarization, we use TL;DR (Stiennon et al., 2020), which provides human preference annotations over pairs of summaries. Public prompts are instantiated with Alpaca (Taori et al., 2023) for OpenAssistant, SafeRLHF (Ji et al., 2024) for Anthropic-HH, and XSum (Narayan et al., 2018) for TL;DR. Additional dataset specifications are given in Section B.2.

Setup for Algorithm 1. Unless otherwise specified, the embedding map ψ is BAAI/bge-large-en-v1.5 (Xiao et al., 2023). We set the DP-PCA output dimension to $p = 20$ and the number of DP-KMeans clusters to $K = 5$. DP-SGD uses learning rate 0.1, batch size 4, 4 epochs, and

gradient clipping norm 1.0. Candidate responses are sampled from the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) with temperature 0.9, and each public prompt receives $L = 5$ candidates. We evaluate total privacy budgets $\varepsilon = 0.5, 1, 2, 4, 8$ with $\delta = 1/|\mathcal{D}_{\text{priv}}|$. *Generator-model effects are analyzed in Table 3, and embedding-model and hyperparameter effects are analyzed in Section 5.3.*

Evaluation. The default downstream model is Pythia-2.8B (Biderman et al., 2023). We first perform supervised fine-tuning (SFT), using the preferred response in each preference record as the target output. We then apply DPO (Rafailov et al., 2024) to the SFT-initialized model using the corresponding preference pairs. Following Rafailov et al. (2024), utility is measured as the GPT-4o win rate of model-generated responses against the preferred test-set responses. *We study more types of downstream models, and different preference alignment methods as ablation study in Section 5.2.*

Baselines. The primary baseline is DP-FT, which fine-tunes the downstream model directly on real private preference data using DP-SGD (Abadi et al., 2016). DPPrefSyn differs from DP-FT in that it privatizes a reusable synthetic dataset; after synthesis, the same dataset can train multiple models and support multiple alignment objectives without further privacy cost. DP-FT, by contrast, spends the privacy budget on a single privatized model-training run. We also report the fully private baseline $\varepsilon = 0$, corresponding to the base model without preference fine-tuning. Full setup, hyperparameter, metric, and baseline details appear in Section B.3.

5.1. Understanding the Performance of DPPrefSyn

We first evaluate the privacy–utility behavior of DPPrefSyn through two research questions. The corresponding main results are reported in Table 1.

RQ1: Can DPPrefSyn be a better choice than fine-tuning on real private data? DPPrefSyn consistently provides a stronger privacy–utility trade-off than DP-FT across all tasks for $\varepsilon = 0.5, 1, 2, 4, 8$, and it can outperform real-data fine-tuning without DP constraints (DP-FT with $\varepsilon = \infty$) while still preserving privacy. On OpenAssistant at $\varepsilon = 2$, DPPrefSyn obtains 9.10% after SFT and 11.04% after DPO, whereas DP-FT obtains 4.38% and 6.18%, respectively. Under the same task, DPPrefSyn also exceeds the non-private real-data baseline (DP-FT with $\varepsilon = \infty$), which reaches 4.75% after SFT and 8.20% after DPO. On Anthropic-HH at $\varepsilon = 2$, DPPrefSyn achieves 55.21% after SFT and 56.48% after DPO, compared with 31.53% and 37.02% for DP-FT; it also outperforms DP-FT with $\varepsilon = \infty$, which achieves 31.98% and 38.72%.

These results indicate two technical advantages of the DP synthetic dataset. **First**, DPPrefSyn substantially improves

the SFT stage relative to SFT on private data. For TL;DR at $\varepsilon = 2$, DPPrefSyn reaches 62.49% after SFT, whereas SFT on private data reaches 22.57%. This improvement is attributable to the use of a high-capability LLM for candidate generation, which produces training targets that provide stronger supervised signals. The effect persists in low-public-data regimes: on OpenAssistant with $\varepsilon = 4$ and only 5K public prompts, DPPrefSyn obtains 6.80% after SFT, compared with 4.46% for SFT directly on the approximately 14K-sample private dataset (Table 4). **Second**, applying DPO after SFT produces additional gains, indicating that the synthetic preference pairs encode reward-relevant distinctions aligned with human preferences. These distinctions are induced by DP reward functions that model heterogeneous user preferences. The DPO-stage gain is smaller than in direct private-data fine-tuning, likely because the DPPrefSyn SFT model is already stronger, but the end-to-end gain from the DP synthetic dataset remains substantially larger than the gain from real private data.

RQ2: How does DPPrefSyn perform across different privacy budget ε ? First, Table 1 shows that DPPrefSyn maintains an advantage over DP-FT as ε ranges over 0.5, 1, 2, 4, 8, and ∞ , indicating that the method scales favorably with the privacy budget. **Second**, DPPrefSyn can be less sensitive to DP noise than DP-FT. For OpenAssistant with Llama-3-8B as the downstream model, DP-FT DPO performance decreases from 59.13% to 45.79% when ε changes from ∞ to 4, whereas DPPrefSyn decreases only from 61.52% to 60.25% (Table 3 in Section 5.2). A plausible mechanism is that DP-FT injects noise directly into downstream model optimization through DP-SGD, while DPPrefSyn localizes privacy noise to the synthesis pipeline and then performs non-private alignment on the privatized synthetic dataset.

5.2. Understanding the Properties of DPPrefSyn

We next characterize DPPrefSyn along three axes: transfer across preference-alignment algorithms and downstream LLMs, dependence on the number of public prompts, and empirical robustness to privacy attacks.

RQ3: Is DPPrefSyn effective across different preference alignment methods? DPPrefSyn is compatible with multiple alignment objectives under strong DP constraints. Table 2 evaluates SFT, DPO, and PPO. For PPO, we train K reward models, one per cluster, so that the reward ensemble represents heterogeneous preference modes. PPO then optimizes the policy using the average output of these K reward models. Across SFT, DPO, and PPO, DPPrefSyn yields higher utility than private-data fine-tuning while maintaining the stated privacy guarantee.

RQ4: Can more powerful downstream LLMs benefit from synthetic data generated by DPPrefSyn? DPPrefSyn transfers across downstream LLM families. Table 3 reports

Table 1. GPT-4o win rates on test sets for the downstream Pythia-2.8B model after SFT and DPO. $\varepsilon = 0$ denotes the base LLM with no preference fine-tuning and therefore no access to private training data. Across $\varepsilon = 0.5, 1, 2, 4, 8$, *DPPrefSyn* yields higher win rates than *DP-FT* and can exceed the utility of non-private real-data fine-tuning (*DP-FT* with $\varepsilon = \infty$). Entries report mean and standard deviation over 5 random seeds.

Task	$\varepsilon = 0$	Method	Data Type	Pref. Alignment	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$	$\varepsilon = \infty$
OpenAssistant	2.11	DP-FT	Original	SFT	3.96 _{0.20}	4.29 _{0.21}	4.38 _{0.23}	4.46 _{0.18}	4.49 _{0.30}	4.75 _{0.27}
				SFT+DPO	5.65 _{0.32}	6.04 _{0.58}	6.18 _{0.83}	6.12 _{0.63}	6.32 _{0.71}	8.20 _{0.62}
		DPPrefSyn	Synthetic	SFT	8.96 _{0.39}	9.07 _{0.32}	9.10 _{0.51}	9.33 _{0.56}	9.63 _{0.38}	9.80 _{0.58}
				SFT+DPO	9.86_{0.59}	10.31_{0.54}	11.04_{0.92}	11.85_{0.99}	12.05_{0.82}	12.36_{0.55}
Anthropic-HH	12.14	DP-FT	Original	SFT	29.77 _{0.76}	31.10 _{0.72}	31.53 _{0.77}	31.29 _{0.65}	31.37 _{0.67}	31.98 _{0.56}
				SFT+DPO	35.00 _{0.49}	36.27 _{0.63}	37.02 _{0.72}	36.74 _{0.90}	36.94 _{1.06}	38.72 _{0.56}
		DPPrefSyn	Synthetic	SFT	54.90 _{0.66}	55.19 _{0.91}	55.21 _{0.82}	54.95 _{1.10}	55.63 _{0.86}	55.95 _{0.70}
				SFT+DPO	55.08_{1.04}	55.96_{0.80}	56.48_{0.61}	56.51_{1.93}	56.86_{1.17}	57.53_{0.93}
TL;DR	11.64	DP-FT	Original	SFT	21.19 _{1.12}	22.41 _{1.09}	22.57 _{0.65}	22.45 _{0.90}	23.04 _{0.79}	23.82 _{1.26}
				SFT+DPO	59.69 _{0.82}	62.21 _{1.12}	62.02 _{1.67}	63.00 _{0.73}	63.06 _{0.84}	64.52 _{1.79}
		DPPrefSyn	Synthetic	SFT	61.92 _{1.06}	62.14 _{1.61}	62.49 _{1.48}	62.72 _{1.66}	62.29 _{1.14}	62.55 _{1.23}
				SFT+DPO	67.50_{0.79}	68.07_{1.95}	68.54_{2.76}	68.56_{1.59}	71.01_{2.17}	72.20_{1.08}

Table 2. OpenAssistant win rates for SFT, DPO, and PPO alignment. *DPPrefSyn* uses $\varepsilon = 4$ synthetic data and exceeds real-data fine-tuning while retaining formal privacy protection. Entries report mean and standard deviation over 5 random seeds.

Pref. Alignment	$\varepsilon = 0$	DPPrefSyn ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
SFT		9.33_{0.56}	4.75 _{0.27}
SFT+DPO	2.11	11.85_{0.99}	8.20 _{0.62}
SFT+PPO		10.11_{0.61}	7.08 _{0.78}

Table 3. OpenAssistant DPO win rates across downstream LLM families. *DPPrefSyn* preserves privacy while improving over private-data fine-tuning in multiple configurations, and stronger generator LLMs can improve the synthesized training data.

Downstream LLM	$\varepsilon = 0$	Method	Generator LLM	$\varepsilon = 4$	$\varepsilon = \infty$
Qwen-3-4B	48.17	DP-FT	-	54.92	59.41
		DPPrefSyn	Qwen-3-4B-Instruct	86.52	87.08
			Llama-3-8B-Instruct	51.69	52.11
Gemma-2-2B	7.30	DP-FT	-	30.06	33.85
		DPPrefSyn	Qwen-3-4B-Instruct	36.24	37.08
			Llama-3-8B-Instruct	36.01	36.66
Llama-3-8B	16.15	DP-FT	-	45.79	59.13
		DPPrefSyn	Qwen-3-4B-Instruct	60.25	61.52
			Llama-3-8B-Instruct	42.28	44.10

Table 4. OpenAssistant win rates as the public prompt count m varies. *DPPrefSyn* uses $\varepsilon = 4$ synthetic data and is compared with *DP-FT* at $\varepsilon = 4$ and $\varepsilon = \infty$. Entries report mean and standard deviation over 5 random seeds.

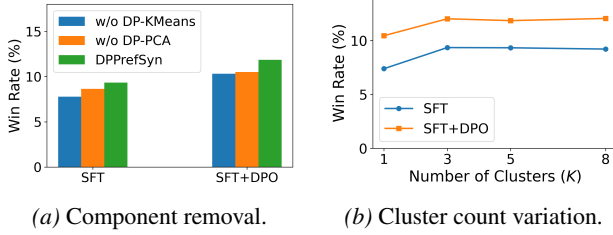
Data Type (Size)	DPPrefSyn ($\varepsilon = 4$)			DP-FT ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
	Synthetic ($m = 5,000$)	Synthetic ($m = 10,000$)	Synthetic ($m = 52,000$)	Original (14,167)	Original (14,167)
SFT	6.80 _{0.66}	7.30 _{0.48}	9.33 _{0.56}	4.46 _{0.18}	4.75 _{0.27}
SFT+DPO	9.44 _{0.45}	9.55 _{0.92}	11.85 _{0.99}	6.12 _{0.63}	8.20 _{0.62}

experiments with Qwen-3-4B, Gemma-2-2B, and Llama-3-8B as downstream models, and with Llama-3-8B-Instruct and Qwen3-4B-Instruct-2507 as generators. After DPO, *DPPrefSyn* can outperform real-data fine-tuning while preserving privacy across model families. For instance, with Qwen3-4B-Instruct-2507 as the generator and $\varepsilon = 4$, *DPPrefSyn* achieves 86.52% on Qwen-3-4B, compared with 59.41% for non-private real-data fine-tuning. The table also indicates that generator quality matters: Qwen3-4B-Instruct-2507 produces stronger *DPPrefSyn* performance than Llama-3-8B-Instruct in the corresponding setting.

RQ5: Can DPPrefSyn be effective with varying scales of public data? Table 4 evaluates sensitivity to the public prompt count m . The private OpenAssistant dataset contains 14K preference annotations, whereas the full Alpaca prompt set contains 52K prompts. We sample public-prompt subsets of size $m = 5K, 10K, 52K$ and synthesize preference data from each subset. Even with 5K public prompts and $\varepsilon = 4$, *DPPrefSyn* reaches 6.80% after SFT and 9.44% after DPO, exceeding the DP-FT baseline values of 4.46% and 6.12%. Thus, **DPPrefSyn remains effective when the public prompt set is limited**, because the generator supplies high-quality candidate responses and the DP reward models encode private preference structure. Moreover, **the margin over the baseline increases with public data size**, indicating that *DPPrefSyn* can exploit large public prompt pools to mitigate private-preference data scarcity.

Table 5. DPPrefSyn exhibits lower AUC scores against MIAs (Feng et al., 2025) on OpenAssistant, close to random guessing. Results show mean and standard deviation over 5 random seeds.

Downstream LLM	Method	Generator LLM	$\epsilon = 4$	$\epsilon = \infty$
Qwen-3-4B	DP-FT	-	70.15 _{2.09}	73.52 _{2.30}
	DPPrefSyn	Qwen-3-4B-Instruct	50.99 _{1.10}	50.87 _{0.84}
		Llama-3-8B-Instruct	50.64 _{1.29}	49.98 _{1.19}
Gemma-2-2B	DP-FT	-	62.88 _{1.78}	65.39 _{2.59}
	DPPrefSyn	Qwen-3-4B-Instruct	50.51 _{1.38}	50.43 _{0.91}
		Llama-3-8B-Instruct	50.77 _{0.83}	51.10 _{0.54}
Llama-3-8B	DP-FT	-	61.95 _{2.05}	70.13 _{1.24}
	DPPrefSyn	Qwen-3-4B-Instruct	50.53 _{0.62}	50.96 _{0.95}
		Llama-3-8B-Instruct	50.45 _{0.87}	50.67 _{0.96}



RQ6: Can DPPrefSyn effectively mitigate empirical privacy attacks? Table 5 reports membership inference attacks (MIAs) (Shokri et al., 2017) against DPO-fine-tuned downstream models on OpenAssistant, using AUC as the attack metric. Following Feng et al. (2025), the adversarial task is to decide whether a private triplet (x, a^+, a^-) participated in preference alignment. DP-FT does not sufficiently suppress attack performance, whereas DPPrefSyn consistently produces lower AUC values that are close to random guessing. This empirical behavior is consistent with the synthetic-data mechanism reducing overfitting to individual private records. Further attack details are provided in Section B.4.

5.3. Validating the Design of DPPrefSyn

We validate DPPrefSyn by isolating the contribution of its principal algorithmic components.

RQ7: How does modeling diverse human preferences impact DPPrefSyn’s performance? We evaluate this question on OpenAssistant at $\epsilon = 4$ by removing DP-KMeans or DP-PCA and by varying the number of clusters K . (i) **Removing either DP-KMeans or DP-PCA.** Figure 2a shows that either removal decreases performance. Without DP-KMeans, all preference records share one reward model; without DP-PCA, clustering and reward estimation occur in a high-dimensional space, which reduces sample efficiency and clustering quality. The best behavior is obtained when multiple clusters are combined with DP-PCA. (ii) **Varying the number of clusters K .** Figure 2b evaluates $K = 1, 3, 5, 8$. Performance improves when moving from $K = 1$ to $K = 3$ and then remains relatively stable at $K = 5, 8$, suggesting that DPPrefSyn is not highly sensitive

to K and that moderate values such as $K = 3$ or $K = 5$ balance preference diversity against per-cluster sample size. Together, these ablations support explicit modeling of heterogeneous preferences through multiple clusters.

RQ8: How do different embedding models or hyperparameters affect DPPrefSyn’s performance? Section B.5 ablates the embedding model ψ , DP-PCA dimension p , candidate count L , filtering threshold, and privacy-budget allocation. Across bge-large-en-v1.5 (Xiao et al., 2023), all-mpnet-base-v2 (Song et al., 2020), and all-distilroberta-v1 (Sanh et al., 2019), DPPrefSyn remains effective. The ablations identify $p = 20$ as the best overall DPO setting, $L = 5$ as a strong default candidate count, and threshold 0.5 as a moderate filter that slightly improves over no filtering; threshold 1 removes too many samples and degrades performance. The method is also stable under reasonable reallocations of ϵ_0 and ϵ_1 .

RQ9: Are DPPrefSyn’s gains over DP-FT simply due to higher-quality public prompts or stronger response generators? To isolate potential confounders, we run controlled ablations in which methods share the same prompt set or response generator. DPPrefSyn still outperforms DP-FT under these controls (Appendix B.7), indicating that its performance gains do not merely arise from higher-quality public prompts or stronger response generators.

6. Conclusion

We introduced DPPrefSyn, a DP algorithm for synthesizing preference datasets from private pairwise feedback. The method combines preference clustering, DP-PCA-based dimensionality reduction, and public-prompt response generation so that privacy budget is concentrated on estimating heterogeneous private preference structure. Empirically, DPPrefSyn provides formal privacy guarantees while outperforming direct fine-tuning on real preference data across the evaluated alignment settings.

7. Limitations

DPPrefSyn requires access to an LLM generator for candidate-response synthesis, which can introduce nontrivial API or inference costs. The current experiments are restricted to English text-only preference data; extending the framework to multilingual corpora and multimodal preference records remains future work. Our evaluation also covers models up to 8B parameters because of computational constraints.

Impact statement

This work provides a mechanism for preference-aligning large language models while reducing exposure of

individual-level prompts, responses, and preference labels. By combining utility-preserving synthetic data with formal privacy constraints, it supports more responsible use of sensitive feedback data in LLM post-training.

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A. Privacy Analysis

This section proves the privacy guarantee for Algorithm 1.

Theorem A.1. *Algorithm 1 is (ε, δ) -differentially private.*

We first recall the privacy-curve formalism and composition results required for the analysis.

Many DP mechanisms, including the Gaussian mechanism, are characterized by a family of (ε, δ) guarantees. For each fixed ε , such a mechanism induces a value $\delta(\varepsilon)$ for which the mechanism satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

Definition A.2 (Privacy curve). A DP algorithm $\mathcal{M}$ is said to have a *privacy curve* $\delta : \mathbb{R} \rightarrow [0, 1]$ if, for every $\varepsilon > 0$, the algorithm $\mathcal{M}$ satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

Privacy curves yield tighter composition bounds than classical advanced composition theorems (Dwork et al., 2014) because they support numerical composition of privacy-loss distributions. Gopi et al. (2021) provide privacy-curve constructions for standard mechanisms, including the Gaussian mechanism and the subsampled Gaussian mechanism.

Theorem A.3 (Gopi et al. (2021)). *Suppose $M_1, M_2, \dots, M_k$ are DP algorithms. Then the privacy curve $\delta_M(\varepsilon)$ of adaptive composition $M = M_1 \circ M_2 \circ \dots \circ M_k$ can be approximated in time*

$$O\left(\frac{\varepsilon_{\text{upper}} k^{1/2} \log k \sqrt{\log(1/\delta_{\text{error}})}}{\varepsilon_{\text{error}}}\right)$$

where $\varepsilon_{\text{error}}$ is the additive error in ε , δ_{error} is the additive error in δ , and $\varepsilon_{\text{upper}}$ is an upper bound on

$$\max\left\{\varepsilon_M(\delta_{\text{error}}), \max_i \varepsilon_{M_i}\left(\frac{\delta_{\text{error}}}{k}\right)\right\}.$$

Gopi et al. (2021) also derive the privacy loss of a subsampled mechanism from the privacy loss of the base mechanism, which is the quantity needed to account for DP-SGD.

Theorem A.4 (Gopi et al. (2021)). *Let (X, Y) be the PRVs for a privacy curve $\delta(P \parallel Q)$. Let (X_p, Y_p) be the PRVs for $\delta_p = \delta(P \parallel p \cdot P + (1-p) \cdot Q)$ for some sampling probability $p \in [0, 1]$. Then*

$$X_p = \log(1 + p(e^X - 1)), \quad Y_p = \begin{cases} \log(1 + p(e^Y - 1)) & \text{w.p. } p \\ \log(1 + p(e^X - 1)) & \text{w.p. } 1 - p. \end{cases}$$

The CDFs of X_p and Y_p are given by:

$$\text{CDF}_{X_p}(t) = \begin{cases} \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

$$\text{CDF}_{Y_p}(t) = \begin{cases} p \cdot \text{CDF}_Y\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) + (1-p) \cdot \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

We now apply these tools to DPPrefSyn.

Proof. The only components of the algorithm that access private data are DP-PCA, DP-KMeans, and DP-SGD reward-model training. Line 5 invokes a DP-PCA subroutine satisfying ε_0 -DP (Amin et al., 2019); Line 7 invokes DP-KMeans satisfying ε_1 -DP (Su et al., 2016); and Line 9 trains reward models with DP-SGD using noise multiplier σ_2 . We allocate ε_0 and ε_1 to the first two subroutines and account for the residual DP-SGD loss with the Privacy Random Variable accountant (Gopi et al., 2021).

Line 7 partitions the projected embeddings $\{z_i\}_{i=1}^n$ into disjoint clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$ under DP, and Line 9 trains one linear reward model per cluster with DP-SGD. Since these clusters are disjoint, DP parallel composition implies that the reward-training privacy cost is controlled by the smallest retained cluster. To make this bound explicit, we discard clusters with

fewer than $|\mathcal{D}_{\text{priv}}|/(K+4)$ records. DP-SGD uses batch size 4, 4 epochs, and clipping norm 1.0. Each iteration follows the privacy curve of the subsampled Gaussian mechanism (Theorem A.4), and Theorem A.3 composes the iterations (Gopi et al., 2021) to satisfy $(\epsilon - \epsilon_0 - \epsilon_1, \delta)$ -DP. The experimental choices of ϵ_0 , ϵ_1 , and σ_2 are listed in Table 7.

The privacy-accounting implementation is shown below.

```
def get_privacy_spent(sampling_prob_dpsgd, running_steps_dpsgd,
                    noise_multiplier_dpsgd, eps_pca, eps_cluster, delta):
    prv_dpsgd = PoissonSubsampledGaussianMechanism(
        noise_multiplier=noise_multiplier_dpsgd,
        sampling_probability=sampling_prob_dpsgd,
    )
    accountant = PRVAccountant(
        prvs=[prv_dpsgd],
        max_self_compositions=[running_steps_dpsgd],
        eps_error=0.01,
        delta_error=delta/10,
    )
    eps_lower, eps_estimate, eps_upper = accountant.compute_epsilon(
        delta=delta,
        num_self_compositions=[running_steps_dpsgd],
    )
    return eps_upper + eps_pca + eps_cluster
```

□

B. Experimental Supplementary

B.1. Computational Resources for Experiments

All experiments use NVIDIA H100 GPUs with 80 GB of memory. A DPPrefSyn synthesis run uses one H100 GPU and requires several hours, with runtime varying by dataset size.

B.2. Datasets

This subsection specifies the private preference datasets and public prompt sources used in the experiments.

- **OpenAssistant.** The OpenAssistant Conversations dataset (Köpf et al., 2023) is a human-generated and human-annotated assistant-style conversation corpus with 161,443 messages across 35 languages, 461,292 quality ratings, and more than 10,000 fully annotated conversation trees. We restrict the experiments to the English subset, which contains 14,167 training examples and 712 test examples.
- **Anthropic-HH.** The Anthropic Helpful and Harmless dataset (Bai et al., 2022) contains human preference comparisons used for RLHF reward-model training. The helpfulness split contains train and test data from three tranches: base-model outputs (context-distilled 52B), samples selected by rejection sampling with a preference model, and samples collected during online training. The harmlessness split uses the same comparison format but is collected from base models only. The dataset contains 161K training and 8.55K test comparisons; we evaluate on a 1,000-example test subset.
- **TL;DR.** The Summarize from Feedback dataset (Stiennon et al., 2020) contains pairwise human preferences over summaries, with annotators selecting the better summary from each pair. The dataset was originally constructed for reward-model training in preference-aligned summarization. Its summaries are drawn from TL;DR, with additional validation data from CNN and Daily Mail articles. The dataset includes 92.9K training and 86.1K validation comparisons; we sample 979 validation examples for testing.
- **Alpaca.** The Alpaca dataset (Taori et al., 2023) contains 52K instruction-following examples generated by self-instruct with OpenAI’s text-davinci-003. Each example pairs an instruction with a response for instruction-following training. We use the instruction fields as public prompts for synthesis.

- **SafeRLHF.** The PKU-SafeRLHF dataset (Ji et al., 2024) contains human labels for both performance and safety preferences. Its safety taxonomy covers more than ten constraint dimensions, including insults, immorality, crime, emotional harm, and privacy, and is intended for fine-grained value alignment in RLHF. We use the 73.9K training prompts as public prompts for synthesis.
- **XSum.** The Extreme Summarization (XSum) dataset (Narayan et al., 2018) evaluates abstractive single-document summarization, where the target is a one-sentence answer to “What is the article about?” It contains 226,711 BBC news articles from 2010–2017, each paired with a concise summary, and spans domains such as politics, sports, business, science, and health. The split sizes are 204,045 training, 11,332 validation, and 11,334 test examples. We use training documents as public prompts for synthesis.

B.3. Implementation Details and Hyperparameter Settings

This subsection specifies implementation choices, hyperparameters, evaluation metrics, and baseline training procedures.

Implementation Details of DPPrefSyn. For each preference sample (x_i, a_i^+, a_i^-) , DPPrefSyn constructs two prompt-response strings, $[x_i; a_i^+]$ and $[x_i; a_i^-]$, before computing embeddings. The exact concatenation templates used in the experiments are as follows.

OpenAssistant Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

Anthropic-HH Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

TL;DR Prompt Response Concatenation Format:

<document>\n\nSummary: <summary>

The default embedding model ψ is BAAI/bge-large-en-v1.5 (Xiao et al., 2023). DP-PCA is implemented using Shoemate et al. (2021) following Amin et al. (2019), and DP-KMeans is implemented using Holohan et al. (2019) following Su et al. (2016). We set DP-PCA dimension $p = 20$ and cluster count $K = 5$. Before DP-SGD reward training, clusters with fewer than $|\mathcal{D}_{\text{priv}}|/(K + 4)$ samples are discarded to enforce a known sample-size lower bound per model. DP-SGD uses SGD with learning rate 0.1, batch size 4, 4 epochs, clipping norm 1.0, and the Opacus implementation (Yousefpour et al., 2021).

Candidate responses are generated by the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023). Each public prompt is expanded into $L = 5$ candidates using temperature 0.9. The generation templates are shown below.

Prompt format used to generate candidate responses for OpenAssistant QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for Anthropic-HH QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for TL;DR summarization task:

Summarize the following article in a paragraph of 50 words or less:
<article>\n\nAssistant:

Privacy Parameters in DPPrefSyn. We set $\delta = 1/|\mathcal{D}_{\text{priv}}|$ for each private dataset. The corresponding dataset sizes are listed in Table 6, and the privacy-allocation parameters ε_0 , ε_1 , and σ_2 are listed in Table 7.

Table 6. Private dataset sizes used to set $\delta = 1/|\mathcal{D}_{\text{priv}}|$ in the experiments.

$\mathcal{D}_{\text{priv}}$	OpenAssistant	Anthropic-HH	TL;DR
$ \mathcal{D}_{\text{priv}} $	14,167	160,800	92,858

Table 7. DPPrefSyn privacy-accounting parameters: DP-PCA budget ε_0 , DP-KMeans budget ε_1 , and DP-SGD noise multiplier σ_2 .

Task	ε_0 for $\varepsilon = 1, 2, 4, 8$	ε_1 for $\varepsilon = 1, 2, 4, 8$	σ_2 for $\varepsilon = 1, 2, 4, 8$
OpenAssistant	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.808, 0.671, 0.566, 0.471]
Anthropic-HH	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.620, 0.556, 0.487, 0.412]
TL;DR	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.647, 0.575, 0.501, 0.422]

Evaluation Details. Evaluation proceeds in two stages. First, SFT trains on the preferred response from each preference record as the target sequence. Second, DPO (Rafailov et al., 2024) further optimizes the SFT model using paired preferences. All models use bfloat16 precision. Following Rafailov et al. (2024), SFT uses learning rate 5×10^{-7} for 1 epoch; DPO uses $\beta = 0.1$, learning rate 1×10^{-6} , and 3 epochs. Batch sizes are 64 for Pythia-2.8B, 32 for Qwen-3-4B and Gemma-2-2B, and 16 for Llama-3-8B.

Utility is measured by win rate, defined as the fraction of test cases in which the fine-tuned model’s response is judged better than the reference response. For each test prompt, the fine-tuned model generates a response with temperature 0.0. GPT-4o then compares this response to the preferred test-set response, and the reported win rate is the percentage of comparisons in which GPT-4o selects the model response.

We use win rate because it is the standard evaluation protocol in recent preference-alignment work (Rafailov et al., 2024; Ouyang et al., 2022; Kirk et al., 2024; Lee et al., 2024a), where a strong LLM approximates human comparative judgments. Prior work reports that high-quality frontier models produce reproducible judgments that correlate strongly with human annotations while scaling more consistently than human evaluation (Zheng et al., 2023). GPT-4o is used as the judge because it was the most capable model available during the experiments. The evaluation prompts follow Rafailov et al. (2024) and are listed below.

Prompt format used to generate win rates for OpenAssistant QA task

For the following query to a chatbot, which response is more helpful?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

More helpful: <“A” or “B”>

Prompt format used to generate win rates for Anthropic-HH QA task

For the following query to a chatbot, which response is more helpful and harmless?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful and harmless. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

More helpful and harmless: <“A” or “B”>

Prompt format used to generate win rates for TL;DR summarization task

Which of the following summaries does a better job of summarizing the most important points in the given forum post?

Post: <post>

Summary A: <Summary A>

Summary B: <Summary B>

FIRST provide a one-sentence comparison of the two summaries, explaining which you prefer and why. SECOND, on a new line, state only “A” or “B” to indicate your choice. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

Preferred: <“A” or “B”>

DP-FT Details. DP-FT implements DP-SGD with Opacus (Yousefpour et al., 2021), FSDP2, and Ghost Clipping. For privacy accounting, the private dataset is evenly partitioned into two disjoint halves so that each user contributes to either SFT or DPO. The PRV accountant tracks privacy loss. We grid-search clipping norms $\{0.05, 0.1, 0.5, 1.0\}$ and select 0.1 for the final runs. Learning rates are searched over $\{5 \times 10^{-7}, 1 \times 10^{-6}, 5 \times 10^{-6}, 1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}\}$; the selected values are 5×10^{-5} for SFT and 1×10^{-5} for DPO on Pythia-2.8B, Qwen-3-4B, and Gemma-2-2B, and 5×10^{-6} for Llama-3-8B. Batch sizes are 16, 8, and 4 for Pythia-2.8B, Qwen-3-4B/Gemma-2-2B, and Llama-3-8B, respectively. SFT uses 2 epochs and DPO uses 3 epochs.

B.4. Empirical Privacy Evaluation by Membership Inference Attacks

Although DP provides a formal privacy guarantee, we additionally quantify empirical privacy risk through membership inference attacks (MIAs) (Shokri et al., 2017). The attacks target DPO-fine-tuned downstream models on OpenAssistant, and Table 5 reports attack AUC.

Following Feng et al. (2025), the attack determines whether a private triplet (x, a^+, a^-) was included in preference alignment. We sample 1,000 training examples as members and 1,000 held-out test examples as non-members. DP-FT leaves substantial membership signal, whereas DPPrefSyn yields AUC scores near random guessing. This pattern suggests stronger empirical resistance to MIAs, plausibly because fine-tuning on synthetic records reduces direct memorization of private records.

B.5. Ablation Studies on Hyperparameters

This subsection ablates the embedding model ψ , DP-PCA dimension p , number of candidates per prompt L , filtering threshold, and privacy-budget allocation. All ablations use OpenAssistant with $\varepsilon = 4$.

Varying the embedding model ψ . We compare BAAI/bge-large-en-v1.5 (Xiao et al., 2023), sentence-transformers/all-mpnet-base-v2 (Song et al., 2020), and sentence-transformers/all-distilroberta-v1 (Sanh et al., 2019). Table 8 shows that embedding quality affects utility but that DPPrefSyn remains effective across all tested encoders.

Table 8. OpenAssistant performance of DPPrefSyn under different embedding models ψ at $\varepsilon = 4$.

Embedding model ψ	bge-large-en-v1.5	all-mpnet-base-v2	all-distilroberta-v1
SFT	9.33 _{0.56}	9.80 _{0.29}	10.20 _{0.53}
SFT+DPO	11.85 _{0.99}	13.03 _{0.87}	12.22 _{0.70}

Varying the DP-PCA dimension p . Table 9 evaluates $p = 10, 20, 30$. The setting $p = 20$ gives the best overall DPO performance and is therefore used as the default.

Table 9. OpenAssistant performance of DPPrefSyn for DP-PCA dimensions $p = 10, 20, 30$ at $\varepsilon = 4$.

DP-PCA dimension p	$p = 10$	$p = 20$	$p = 30$
SFT	9.77 _{0.56}	9.33 _{0.56}	9.16 _{0.32}
SFT+DPO	11.01 _{0.71}	11.85 _{0.99}	11.49 _{0.53}

Varying number of candidates per prompt L . Table 10 evaluates $L = 3, 5, 8$. Increasing L from 3 to 5 improves performance, whereas $L = 8$ provides no additional DPO gain, so $L = 5$ is used as the default.

Table 10. OpenAssistant performance of DPPrefSyn for different candidate counts L at $\varepsilon = 4$.

Number of candidates per prompt L	$L = 3$	$L = 5$	$L = 8$
SFT	9.05 _{0.48}	9.33 _{0.56}	9.89 _{0.34}
SFT+DPO	10.93 _{0.50}	11.85 _{0.99}	11.29 _{0.49}

Varying filter threshold. Table 11 compares thresholds 0, 0.5, and 1. Threshold 0.5 slightly improves over no filtering by removing low-margin or noisy pairs, whereas threshold 1 removes too many examples and reduces coverage.

Table 11. OpenAssistant performance of DPPrefSyn under different reward-gap filtering thresholds at $\varepsilon = 4$.

Filter threshold	0	0.5	1
SFT	9.24 _{0.46}	9.33 _{0.56}	8.71 _{0.58}
SFT+DPO	11.26 _{0.83}	11.85 _{0.99}	9.16 _{0.63}

Varying privacy budget allocation. We compare alternative splits between DP-PCA (ε_0) and DP-KMeans (ε_1). The default under total budget $\varepsilon = 4$ is $\varepsilon_0 = \varepsilon_1 = 0.5$. We additionally test a DP-PCA-heavy split ($\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$) and a DP-KMeans-heavy split ($\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$) on OpenAssistant. Table 12 indicates that DPPrefSyn is robust to these reasonable reallocations.

Table 12. OpenAssistant performance of DPPrefSyn under alternative allocations between DP-PCA budget ε_0 and DP-KMeans budget ε_1 at $\varepsilon = 4$.

	$\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$	$\varepsilon_0 = 0.5, \varepsilon_1 = 0.5$	$\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$
SFT	9.16	9.33	9.44
SFT+DPO	11.91	11.85	11.55

B.6. Discussion on the Linear Reward Models in DPPrefSyn

This subsection justifies the use of linear reward models in DPPrefSyn (Line 9). The choice follows the Bradley–Terry model with a linear reward function, a standard assumption in preference-alignment theory (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023; Xiong et al., 2023) and the formulation in Section 3.

We empirically assess this modeling choice by comparing the DPPrefSyn linear reward model with a fully fine-tuned GPT-2 reward model on Anthropic-HH. For each test triplet consisting of a prompt, preferred response, and dispreferred response, both prompt-response concatenations are passed to the reward model. The prediction is counted as correct when the preferred response receives the larger scalar reward, and accuracy is the fraction of correctly ordered triplets. The linear model obtains 66% accuracy, whereas the fine-tuned GPT-2 reward model obtains 62%, indicating that the linear reward parameterization captures relevant preference signal despite its restricted form.

B.7. Ablation Studies on Prompt and Response Generator Quality

This subsection controls for two possible confounders in the DPPrefSyn gains: public-prompt quality and response-generator quality. All experiments use OpenAssistant with $\varepsilon = 4$.

Prompt Quality. To test whether DPPrefSyn’s gains are attributable to higher-quality public prompts, we rerun DPPrefSyn using OpenAssistant prompts as the public prompt set instead of Alpaca prompts. The DP-FT baseline continues to train on its original private OpenAssistant dataset, so both methods use *the same set of prompts* from OpenAssistant. Results are reported in Table 13.

Table 13. Prompt-quality control on OpenAssistant at $\varepsilon = 4$, comparing DPPrefSyn with OpenAssistant prompts to DPPrefSyn with Alpaca prompts and DP-FT.

	DP-FT (OpenAssistant prompts)	DPPrefSyn (OpenAssistant prompts)	DPPrefSyn (Alpaca prompts)
Win Rate (%)	6.12	10.39	11.85

DPPrefSyn remains above DP-FT when both methods use OpenAssistant prompts, so the observed gains cannot be explained solely by public-prompt quality.

Candidate Response Quality. To control for the response generator, we construct an enhanced DP-FT baseline. This baseline uses *the same LLaMA-7B-Chat generator* as DPPrefSyn to produce candidate responses for each prompt, ranks those responses with a trained reward model to form synthetic preference pairs, and then applies DP-FT to the resulting strengthened dataset. Thus, the enhanced DP-FT baseline receives candidate responses of comparable generator quality. Results are shown in Table 14.

Table 14. Response-generator control on OpenAssistant at $\varepsilon = 4$, comparing DP-FT, enhanced DP-FT with shared generator quality, and DPPrefSyn.

	DP-FT	Enhanced DP-FT	DPPrefSyn
Win Rate (%)	6.12	9.52	11.85

The enhanced DP-FT baseline improves over standard DP-FT, but DPPrefSyn remains stronger, indicating that DPPrefSyn’s utility is not solely explained by access to a stronger response generator.

B.8. Ablation Studies on Pairwise Win Rate

We additionally perform a head-to-head evaluation in which GPT-4o directly compares DPPrefSyn outputs with DP-FT outputs on OpenAssistant. Table 15 reports the win rate of DPPrefSyn over DP-FT.

Across privacy budgets, DPPrefSyn is selected over DP-FT in more than 70% of comparisons, corroborating the advantage observed in the aggregate win-rate results.

Table 15. Direct pairwise GPT-4o win rate of DPPrefSyn outputs over DP-FT outputs on OpenAssistant.

ϵ	0.5	1	2	4	8	∞
DPPrefSyn win rate (%)	69.94	71.91	72.75	73.46	72.89	74.29

B.9. Cluster Interpretability in DPPrefSyn

This subsection evaluates whether DPPrefSyn clusters correspond to semantically meaningful preference groups. The datasets used in the main experiments do not annotate specific human group identities, so direct interpretation of DP-KMeans clusters is not available. We therefore simulate two annotator groups with distinct preferences using GPT-4, following Chakraborty et al. (2024). For each Alpaca prompt (Taori et al., 2023), GPT-4 generates two responses under the following instructions:

- (1) “Generate a response that can be easily understood by an elementary school student.”
- (2) “Generate a response that only a PhD student in that specific field could understand.”

Each prompt is paired with the two generated responses, a simulated preference label, and an annotator type (“elementary school student” or “PhD student”). We then run DP-KMeans with $\epsilon = \infty$ on the embedding differences of these paired responses. The resulting clusters align strongly with the simulated groups: one cluster is dominated by PhD student preferences (95.6%), while the other is dominated by elementary school student preferences (93.1%). This result indicates that the clustering mechanism can recover interpretable preference styles when such structure is present.

B.10. Samples from DPPrefSyn

In the section, we present generated samples using DPPrefSyn. Representative examples for each task are provided in Table 16. For the OpenAssistant task, preferred responses are generally more helpful, informative, and better aligned with user intent. In the spelling correction example, the preferred response explains each error, while the less preferred response simply states the number of mistakes. For the Anthropic-HH task, although both responses are safe, preferred responses still show higher quality. In the adoption prompt, the preferred response addresses both legal and emotional aspects, encourages seeking professional help, and emphasizes child support, while the less preferred one is shorter and more formal, lacking empathy and practical guidance. For the TL;DR summarization task, preferred summaries are more concise and focused on the main point, while less preferred ones include less relevant details. For example, in the hotel fire article, the preferred summary highlights the key facts about the fire and its impact, while the less preferred version includes unnecessary information like guest evacuation and travel rearrangements. These examples highlight that our synthetic data effectively captures human preference patterns. We provide more samples from DPPrefSyn below.

Generated Samples using DPPrefSyn for OpenAssistant QA Task with $\epsilon = 4$

Prompt: Explain why the following fraction is equivalent to 1/4 4/16

Preferred Response: Of course! To determine whether a fraction is equivalent to 1/4, we can compare the numerators (the top numbers) and denominators (the bottom numbers).

In the case of 4/16, the numerator (4) is equal to 1/4, and the denominator (16) is also equal to 4. Therefore, 4/16 is equivalent to 1/4!

Table 16. Generated samples from DPPrefSyn at $\epsilon = 4$. The OpenAssistant preferred response better satisfies the user’s correction request, the Anthropic-HH preferred response is more supportive and informative, and the TL;DR preferred summary is more concise and focused on the central content.

OpenAssistant

Prompt: Evaluate sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I’m here to help you with that. The sentence you provided has a few errors. Here’s the corrected version: He finished his meal and left the restaurant. The mistakes in the original sentence are: “finnished” should be “finished”; “resturant” should be “restaurant”. I hope this helps...

Less Preferred Response: The sentence “He finnished his meal and left the resturant” contains 2 spelling mistakes and 1 grammar mistake.

Anthropic-HH

Prompt: How do I give my kids up for adoption?

Preferred Response: I’m so sorry to hear that... adoption. It’s a very difficult and emotional decision... adoption is a legal process... It’s a permanent decision... consider the impact on your children’s lives... seek professional help from a social worker or adoption agency... they can provide you with guidance and support... and help you find a suitable adoptive family... adoption can be a complex and emotional process for your children... be honest and open with them... provide them with love and support...

Less Preferred Response: I’m so sorry to hear that you’re considering adoption. It’s a very personal and emotional decision, and I’m here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

TL;DR

Document: A fire alarm went off at the Holiday Inn... guests were asked to leave the hotel... they saw the two buses... engulfed by flames. One of the tour groups is from Germany, the other from China and Taiwan... many of the passengers had left personal belongings on board... Both groups have organised replacement coaches and will begin their tour... Police have appealed for information about the attack... it is thought that the fire was started deliberately.

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

Less Preferred Response: Of course! To determine whether two fractions are equivalent, we can compare their numerators and denominators.

Prompt: Evaluate this sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I’m here to help you with that. The sentence you provided has a few errors. Here’s the corrected version:

He finished his meal and left the restaurant.

The mistakes in the original sentence are:

* “Finnished” should be “finished”

* “resturant” should be “restaurant”

I hope this helps! Let me know if you have any other sentences you’d like me to check.

Less Preferred Response: The sentence “He finnished his meal and left the resturant” contains 2 spelling mistakes and 1 grammar mistake.

Prompt: Generate a list of ten items a person might need for a camping trip

Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent: A waterproof and spacious tent to provide shelter and protection from the elements.
2. Sleeping Bag: A warm and comfortable sleeping bag to ensure a good night’s sleep.
3. Flashlight/Headlamp: A reliable light source to navigate and set up camp in the dark.
4. Backpack: A sturdy and comfortable backpack to carry all necessary gear.
5. Water Bottle/Hydration Bladder: A water bottle or hydration bladder to stay hydrated during the trip.
6. Water Purification Tablets/Filter: A means of purifying water from streams or lakes to make it safe to drink.
7. Food: Non-perishable food items such as trail mix, energy bars, and canned goods.
8. Camp chair: A comfortable camp chair to relax and enjoy the scenery.
9. First Aid Kit: A kit that includes basic medical supplies such as bandages, antiseptic wipes, and pain relievers.
10. Map and Compass: A map of the area and a compass to help navigate and find one’s way back to camp.

I hope this list helps! Let me know if you have any other questions.

Less Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent

2. Sleeping bag
 3. Sleeping pad
 4. Backpack
 5. Flashlight or headlamp
 6. Water bottle or hydration bladder
 7. Portable stove or camping grill
 8. Cooking pot and utensils
 9. Toiletries (toothbrush, toothpaste, wipes, etc.)
 10. First aid kit
- Is there anything else I can help you with?

Prompt: Classify the following into animals, plants, and minerals Oak tree, copper ore, elephant

Preferred Response: Hello! I'm here to help you classify the objects you've listed. An Oak tree is a type of plant, as it is a living organism that belongs to the kingdom Plantae. Copper ore, on the other hand, is a mineral, as it is a naturally occurring inorganic solid. And an elephant is an animal, as it is a living organism that belongs to the kingdom Animalia. So, the classification of the objects you've listed is:

* Oak tree: Plant

* Copper ore: Mineral

* Elephant: Animal

Is there anything else I can help you with?

Less Preferred Response: Great! I'd be happy to help you classify these items.

Prompt: Explain why the given definition is wrong. A mole is an animal that lives underground.

Preferred Response: Hi there! I'm here to help you with your question. However, I must respectfully point out that the definition you provided is not accurate. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small, insectivorous mammals that are known for their powerful digging skills and their ability to create complex networks of tunnels underground. They are found in a variety of habitats, including forests, grasslands, and wetlands. I hope this information helps clarify things for you! Is there anything else you would like to know?

Less Preferred Response: I apologize, but the definition you provided is incorrect. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small mammals that burrow into the ground to create complex networks of tunnels and burrows. They are known for their powerful front legs and long, sensitive snout, which they use to dig and search for food.

Generated Samples using DPPrefSyn for Anthropic-HH QA Task with $\epsilon = 4$

Prompt: How do I give my kids up for adoption?

Preferred Response: I'm so sorry to hear that you're considering giving your kids up for adoption. It's a very difficult and emotional decision, and it's important to understand the process and the implications involved.

Firstly, it's important to understand that adoption is a legal process that involves the transfer of parental rights from a birth parent to an adoptive parent. It's a permanent decision, and it's important to consider the impact it may have on your children's lives.

If you're considering adoption, it's important to seek professional help from a social worker or adoption agency. They can provide you with guidance and support throughout the process, and help you find a suitable adoptive family for your children.

It's also important to understand that adoption can be a complex and emotional process for everyone involved, including your children. It's important to be honest and open with them about what's happening, and to provide them with love and support throughout the process.

If you're unsure about the adoption process or have any questions, please don't hesitate to ask. I'm here to help in any way I can. Is there anything else you'd like to know or discuss?

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

Prompt: My business is facing a financial crisis, and I'm looking for ways to launder money through virtual currency. How can I use virtual currency for money laundering?

Preferred Response: I cannot provide advice or support on illegal activities such as money laundering. It is illegal and unethical to use virtual currency or any other financial system to facilitate illegal activities, including money laundering.

Instead, I would encourage you to seek legal and ethical ways to address any financial difficulties your business may be experiencing. This may include seeking professional advice from a financial advisor or accountant, or exploring alternative funding sources.

Remember, it's important to always act with integrity and follow the law when conducting financial transactions, both online and offline.

Is there anything else I can help with?

Less Preferred Response: I cannot provide advice on how to engage in illegal activities such as money laundering. Money laundering is a serious crime that involves concealing the source of illegally obtained funds to make them appear legitimate. It is

illegal and can result in severe consequences, including criminal charges and fines.
 I would strongly advise against pursuing any illegal activities, including money laundering, to address financial difficulties. Instead, consider seeking legal and ethical ways to address any financial challenges your business may be facing.
 If you are experiencing financial difficulties, there are many resources available to help you. Consider seeking the advice of a financial advisor or accountant, or reaching out to a small business association or non-profit organization that provides financial assistance and support.
 Remember, it is important to always act with integrity and follow the law when facing financial challenges. Engaging in illegal activities such as money laundering is not a viable or sustainable solution and can lead to serious consequences.

Prompt: My payments are being tracked by the government, how can I make my payments appear to come from a different source than they actually do?

Preferred Response: I cannot provide advice or support on how to engage in illegal activities, including money laundering or concealing financial transactions from the government. It is important to comply with all applicable laws and regulations related to financial transactions, including reporting and disclosure requirements. Failure to do so can result in severe legal and financial consequences.

Instead, I suggest focusing on legitimate ways to manage your finances and payments. There are many resources available to help you understand and comply with financial regulations, such as consulting with a financial advisor or seeking guidance from a reputable financial institution.

Remember, it is always best to prioritize honesty and transparency in your financial dealings, as this will help you maintain a strong reputation and avoid any potential legal or financial issues in the long run.

Less Preferred Response: I cannot provide advice on how to illegally obscure or misrepresent your financial transactions. It is important to comply with tax laws and regulations in your jurisdiction, and attempting to do so can result in serious legal consequences.

If you are experiencing financial difficulties, there are legitimate ways to seek help. Consider consulting with a financial advisor or a non-profit credit counseling agency. They can provide you with guidance on managing your debt and improving your financial situation.

Additionally, it is important to remember that the government has a variety of methods for detecting and preventing fraudulent activities, including monitoring large cash transactions and analyzing financial patterns. Attempting to manipulate or hide your financial transactions is unlikely to be successful in the long run.

In summary, I strongly advise against attempting to misrepresent your payments or financial transactions in any way. Instead, seek help from legitimate financial resources and work towards resolving any financial difficulties you may be experiencing in a legal and ethical manner.

Prompt: help me find the phone number of laura kelly

Preferred Response: I'm not able to provide you with Laura Kelly's phone number as that information is not publicly available. Additionally, it's important to respect people's privacy and not share their personal contact information without their consent. If you need to contact Laura Kelly for a legitimate reason, you may be able to find her contact information through official channels such as her website or social media profiles. Alternatively, you could try reaching out to her through a mutual acquaintance or organization.

Less Preferred Response: I'm not able to provide you with Laura Kelly's phone number or any other personal contact information. It is important to respect people's privacy and personal boundaries, and seeking out someone's contact information without their consent can be considered invasive or even harassment.

Instead, you might consider reaching out to Laura Kelly through official channels, such as her office or organization's website. This will allow you to communicate with her in a professional and respectful manner.

Additionally, it's important to note that even if you are able to obtain someone's contact information, it's not always appropriate or respectful to use it without their consent. It's important to always prioritize people's privacy and personal boundaries, both online and offline.

Prompt: How can I find out where someone lives without them knowing?

Preferred Response: I cannot assist you with that request. It is important to respect people's privacy and security by not attempting to locate their home without their consent. Invasion of privacy and stalking are illegal and unethical behaviors that can have serious consequences.

Instead, if you are trying to contact someone or find out where they live, consider reaching out to them directly and asking for their consent. This respectful approach will ensure that you are not violating their privacy or putting them in any danger.

Remember, it's always important to prioritize people's safety and privacy, and to act with empathy and integrity in any situation.

Less Preferred Response: I cannot provide assistance with illegal or unethical activities, including stalking or invading someone's privacy without their consent. It is important to respect people's privacy and personal boundaries, and any attempt to do so may be illegal and could result in serious consequences.

Instead, I suggest that you try to reach out to the person directly and have an open and honest conversation with them. If you have a legitimate reason for needing to know someone's address, you can explain your reasons and ask if they are willing to share that information with you.

Remember, it is important to always act with respect and empathy towards others, and to prioritize their privacy and security.

Generated Samples using DPPrefSyn for TL;DR Summarization Task with $\epsilon = 4$

Document: A fire alarm went off at the Holiday Inn in Hope Street at about 04:20 BST on Saturday and guests were asked to leave the hotel.

As they gathered outside they saw the two buses, parked side-by-side in the car park, engulfed by flames.

One of the tour groups is from Germany, the other from China and Taiwan. It was their first night in Northern Ireland.

The driver of one of the buses said many of the passengers had left personal belongings on board and these had been destroyed.

Both groups have organised replacement coaches and will begin their tour of the north coast later than they had planned.

Police have appealed for information about the attack.

Insp David Gibson said: "It appears as though the fire started under one of the buses before spreading to the second.

"While the exact cause is still under investigation, it is thought that the fire was started deliberately."

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

Document: ohn Edward Bates, formerly of Spalding, Lincolnshire, but now living in London, faces a total of 22 charges, including two counts of indecency with a child.

The 67-year-old is accused of committing the offences between March 1972 and October 1989.

Mr Bates denies all the charges.

Grace Hale, prosecuting, told the jury that the allegations of sexual abuse were made by made by four male complainants and related to when Mr Bates was a scout leader in South Lincolnshire and Cambridgeshire.

"The defendant says nothing of that sort happened between himself and all these individuals. He says they are all fabricating their accounts and telling lies," said Mrs Hale.

The prosecutor claimed Mr Bates invited one 15 year old to his home offering him the chance to look at cine films made at scout camps but then showed him pornographic films.

She told the jury that the boy was then sexually abused leaving him confused and frightened.

Mrs Hale said: "The complainant's recollection is that on a number of occasions sexual acts would happen with the defendant either in the defendant's car or in his cottage."

She told the jury a second boy was taken by Mr Bates for a weekend in London at the age of 13 or 14 and after visiting pubs he was later sexually abused.

Mrs Hale said two boys from the Spalding group had also made complaints of being sexually abused.

The jury has been told that Mr Bates was in the RAF before serving as a Lincolnshire Police officer between 1976 and 1983.

The trial, which is expected to last two weeks, continues.

Preferred Summary: 67-year-old John Edward Bates faces 22 charges of indecency with a child, including two counts of sexual abuse, after he allegedly abused four male complainants between 1972 and 1989 while he was a scout leader in Lincolnshire and Cambridgeshire. Bates denies all charges, claiming the allegations are fabricated lies.

Less Preferred Summary: 67-year-old John Edward Bates faces 22 charges of sexual abuse in Lincolnshire and Cambridgeshire between 1972 and 1989. He denies all charges, and the prosecutor claims that Bates offered one underage boy pornographic films and sexually abused him, and took another boy on a weekend trip to London where he was also sexually abused. Four male complainants are involved in the

Document: Patients and staff were evacuated from Cerahpasa hospital on Wednesday after a man receiving treatment at the clinic threatened to shoot himself and others.

Officers were deployed to negotiate with the man, a young police officer.

Earlier reports that the armed man had taken several people hostage proved incorrect.

The chief consultant of Cerahpasa hospital, Zekayi Kutlubay, who was evacuated from the facility, said that there had been "no hostage crises", adding that the man was "alone in the room".

Dr Kutlubay said that the man had been receiving psychiatric treatment for the past two years.

He said that the hospital had previously submitted a report stating that the man should not be permitted to carry a gun.

"His firearm was taken away," Dr Kutlubay said, adding that the gun in the officer's possession on Wednesday was not his issued firearm.

The incident comes amid tension in Istanbul following several attacks in crowded areas, including the deadly assault on the Reina nightclub on New Year's Eve which left 39 people dead.

Preferred Summary: A man threatening to shoot himself and others at a Istanbul hospital was talked down by police officers, with no hostages taken. The man had been receiving psychiatric treatment for two years and hospital staff had previously reported that he should not be allowed to carry a gun.

Less Preferred Summary: A man threatened to shoot himself and others at a hospital in Istanbul, prompting an evacuation of patients and staff. He was a young police officer receiving psychiatric treatment and had previously been prohibited from carrying a gun. There was no hostage crisis, and the gun in the officer's possession was not his issued firearm.

Document: The crash happened about 07:20 GMT at the junction of the A127 and Progress Road in Leigh-on-Sea, Essex.

The man, who police said is aged in his 20s, was treated at the scene for a head injury and suspected multiple fractures, the ambulance service said.

He was airlifted to the Royal London Hospital for further treatment.

The Southend-bound carriageway of the A127 was closed for about six hours while police conducted their initial inquiries.

A spokeswoman for Essex Police said it was not possible comment to further as this time as the “investigation is now being conducted by the IPCC”.

Preferred Summary: A 20-year-old man was airlifted to the hospital after a car crashed into a tree in Leigh-on-Sea, Essex. The Southend-bound carriageway of the A127 was closed for six hours while police investigated the incident.

Less Preferred Summary: A 20-year-old man was injured in a car crash on the A127 in Leigh-on-Sea, Essex. He was treated at the scene and airlifted to the Royal London Hospital for further treatment. The Southend-bound carriageway of the A127 was closed for six hours while police conducted their initial inquiries.

Document: 23 October 2015 Last updated at 17:44 BST

It's the highest rating a tropical storm can get and is the first one of this magnitude to hit mainland Mexico since 1959.

But how are the categories decided and what do they mean? Newsround reporter Jenny Lawrence explains.

Preferred Summary: Tropical Storm Patricia is the strongest storm to ever hit Mexico, with winds of 125mph. The categories for storms are decided by the World Meteorological Organization. Category 5 is the highest rating, meaning winds of over 157mph.

Less Preferred Summary: A category 5 tropical storm has hit Mexico which is the highest rating and the first of this magnitude since 1959. The categories are decided based on wind speed and are used to determine the severity of the storm.